Day 3: Kinematics III: Transverse Spectra, Feynman Scaling, and Radial Flow

14-Day Hands-on Course on Heavy-Ion Phenomenology

Lecture

NIT-J, HEP-119

June 2026

Syllabus: 14-Day Hands-on Course Layout

Part I: Relativistic Kinematics

- Day 1: Foundations of Relativistic Kinematics & Experimental HEP
- Day 2: Kinematics I & II: Rapidity & Pseudorapidity in Detail
- **Day 3: Kinematics III: Transverse Spectra, Feynman Scaling & Radial Flow**
- Day 4: Kinematics IV: Coordinate Calculator implementations

Part II: AMPT Physics & Mechanics

- Day 5: AMPT Evolution: HIJING, ZPC, Hadronization, ART
- Day 6: AMPT Data format: event headers & parsing
- Day 7: Single-Particle Observables: Multiplicity & p_T

Part III: Thermodynamics & Collectivity

- Day 8: Transverse Spectra, Identified Particles & m_T Scaling
- Day 9: Centrality & Geometry: Glauber vs AMPT
- Day 10: Collective Flow: Elliptic Flow (v_2)
- Day 11: Medium Modification: ZPC Knobs & Mean Free Path

Part IV: Advanced Scans & Pipeline

- Day 12: Energy Scan: Beam Energy Scan (STAR at RHIC)
- Day 13: Advanced Analysis
- Day 14: Wrap-Up: Reproducible HEP Data Pipeline

Luminosity in Collider Experiments: Theory

Theoretical Definition (Sahoo Eq. 5.188)

The **Luminosity** ($\mathcal{L}$) of a collider characterizes the number of collisions that can be produced per unit area and time. For head-on symmetric beams:

$$\mathcal{L} = \frac{f n_b N_p^2}{4\pi \sigma_x \sigma_y}$$

Where:

- f: Revolution frequency of the bunches.
- n_b: Number of particle bunches per beam.
- N_p: Number of particles per bunch (bunch intensity).
- σ_x, σ_y: Transverse beam profile sizes at the Interaction Point (IP).

Luminosity in Collider Experiments: Beam Squeezing

The Role of Beam Size

According to the luminosity equation $\mathcal{L} \propto \frac{1}{\sigma_x \sigma_y}$, focusing the particles into a smaller transverse area dramatically increases the collision probability density.

Squeezing the Beams with Quadrupoles

Accelerators use specialized electromagnetic quadrupole magnets located immediately upstream of the collision point to focus/squeeze the beams:

- This focusing is characterized by the **beta-star** (β^*) parameter, representing the beam envelope width at the interaction point.
- Squeezing the beam (reducing β^* and thus $\sigma_{x,y}$) is the primary mechanism for maximizing instantaneous luminosity at facilities like the LHC.

Luminosity in Collider Experiments: Detector Meaning

Detector Event Rates & Integrated Luminosity

In detector physics, the event production rate $\mathrm{d}R/\mathrm{d}t$ for a specific process is governed by the accelerator luminosity and the physical cross-section σ:

$$\frac{\mathrm{d}R}{\mathrm{d}t} = \mathcal{L} \cdot \sigma$$

The total accumulated count of events N over time is:

$$N = L \cdot \sigma = \left(\int \mathcal{L} \, \mathrm{d}t \right) \cdot \sigma$$

where L is the **integrated luminosity** (usually measured in inverse picobarns, pb^{-1}).

Interactive Detector Simulator

Open in your browser:

- `animations/11_detector_luminosit`

Observe how bunch frequencies and focusing parameters scale instantaneous event rates, and why rare Higgs events require huge integrated luminosity.

Fixed-Target vs. Collider: Kinematic Advantage

Energy Availability in Collisions

The total center-of-mass energy $\sqrt{s}$ dictates the energy available for new particle production:

- **Fixed-Target Configurations**:

$$\sqrt{s_{\mathrm{FT}}} = \sqrt{m_p^2 + m_t^2 + 2E_{\mathrm{lab}}m_t} \approx \sqrt{2E_{\mathrm{lab}}m_t}$$

  Much of the incoming beam energy is wasted as the center-of-mass frame itself recoils forward at high velocity. $\sqrt{s}$ scales slowly ($\propto \sqrt{E_{\mathrm{lab}}}$).

- **Collider Configurations**:

$$\sqrt{s_{\mathrm{coll}}} = 2E_{\mathrm{lab}}$$

  Two beams collide head-on. The net momentum is zero, meaning the center-of-mass frame is stationary. **All beam energy is available for particle creation**. $\sqrt{s}$ grows linearly with beam energy ($\propto E_{\mathrm{lab}}$).

Fixed-Target vs. Collider: Design Trade-offs

Facility Comparison (Sahoo Table 5.5)

No.	Fixed-Target	Collider
1.	Energy scaling is slow ($\sqrt{s} \propto \sqrt{E_{\rm lab}}$)	Energy grows linearly ($\sqrt{s} \propto E_{\rm cm}$)
2.	Secondary unstable beams (π, K, μ) can be targets	Only stable/charged beams (p, A, e^-, e^+)
3.	High luminosity easily achieved (high target density)	Low interaction rate (diffuse beam-beam density)

Problem 1 in Lab

You will calculate the LHC luminosity ($\mathcal{L} \approx 10^{34}\ \mathrm{cm}^{-2}\mathrm{s}^{-1}$) and compare fixed-target vs. collider runtime requirements for collecting identical statistics.

The Feynman Scaling Variable x_F

Feynman-x Definition (Sahoo Eq. 5.192)

The **Feynman-x variable** (x_F) represents the fraction of the maximum longitudinal momentum carried by a produced particle in the center-of-mass system:

$$x_F = \frac{p_L^*}{p_L^*(\max)} \approx \frac{2p_z^*}{\sqrt{s}}$$

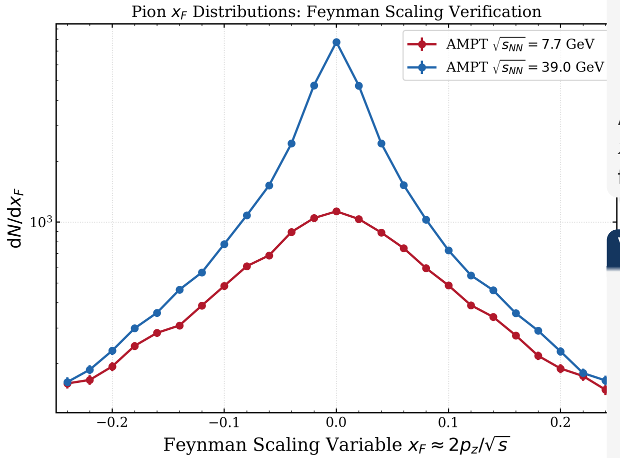
Where p_z^* is the longitudinal momentum in the center-of-mass frame, and $\sqrt{s}$ is the total collision energy.

Feynman Scaling Hypothesis

Feynman hypothesized that at high energies ($\sqrt{s} \to \infty$), the inclusive single-particle distribution functions $f_i(p_T, x_F)$ become independent of the collision energy. This reflects the scale-invariance of the underlying QCD parton distribution functions.

Longitudinal Phase Space Simulator

AMPT: Feynman Scaling & Rapidity Plateau



Feynman-x and Rapidity connection

$$x_F = \frac{m_T}{\sqrt{s}} \sinh y^* \implies y^* = \sinh^{-1}\left(\frac{x_F\sqrt{s}}{m_T}\right)$$

At high energies ($\sqrt{s} \to \infty$), a small finite range of x_F around 0 maps to a very large rapidity interval, forming the central **rapidity plateau**.

Verification with AMPT Data

- Plotting $\mathrm{d}N/\mathrm{d}x_F$ at $\sqrt{s_{NN}} = 7.7$ and 39 GeV shows scaling overlap at higher $|x_F|$ values (fragmentation region).
- The divergence at $x_F \approx 0$ reflects the logarithmic growth of the central rapidity plateau height as energy increases.

Transverse Momentum (p_T) & Mass (m_T) Fit Regimes

Invariant Yield (Sahoo Eq. 5.207)

$$E\frac{\mathrm{d}^3N}{\mathrm{d}p^3} = \frac{1}{2\pi p_T}\frac{\mathrm{d}^2N}{\mathrm{d}p_T\mathrm{d}y} = \frac{1}{2\pi m_T}\frac{\mathrm{d}^2N}{\mathrm{d}m_T\mathrm{d}y}$$

Low-p_T Thermal Regime

$$\text{Yield} \propto e^{-m_T/T_{\mathit{eff}}}$$

Describes particle production from a thermalized bulk medium at effective temperature T_{eff}.

High-p_T Hard Scattering Regime

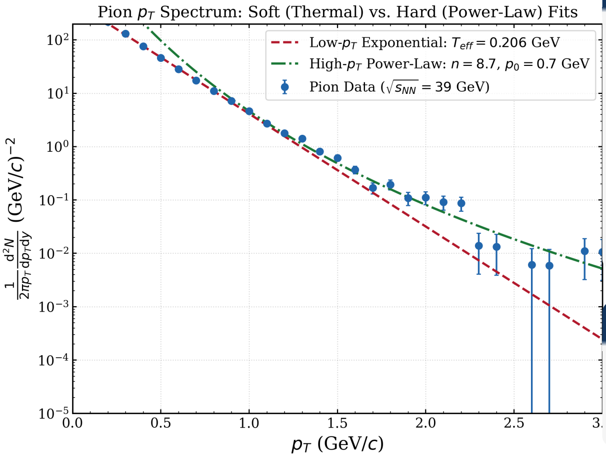
$$\text{Yield} \propto B\left(1+\frac{p_T}{p_0}\right)^{-n}$$

Governed by hard parton scatterings and jet fragmentation described by perturbative QCD.

Interactive Spectra Simulator

Open `animations/12_pt_thermal_powerlaw.html` to visualize the transition from soft thermal bulk emission to hard scattering tails.

AMPT: Soft vs. Hard Fit Regions



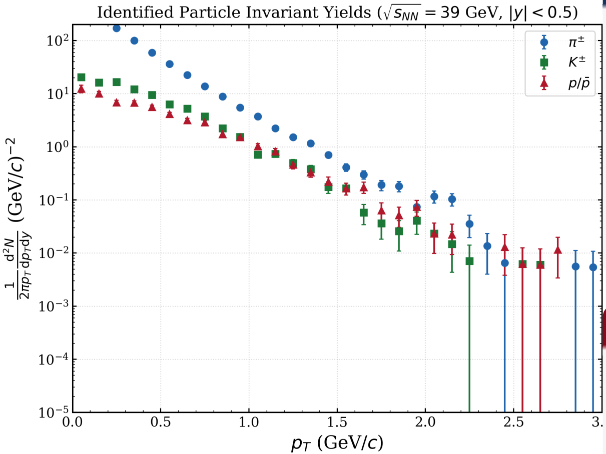
Observations in AMPT Pion Spectrum

- **Low-p_T region ($p_T < 0.8$ GeV/c)**: Follows the exponential thermal shape. Fitting yields $T_{eff} \approx 139$ MeV.
- **High-p_T tail ($p_T > 1.5$ GeV/c)**: The thermal curve drops off exponentially (underestimates statistics by orders of magnitude). The power-law fit describes the hard tail correctly ($n \approx 9.2$).

Physical Mechanism

Soft particle production is statistical/thermal; hard particle production is dynamical (parton-parton collisions).

AMPT: Identified Hadrons Invariant Yields vs. p_T



Transverse Momentum Distribution ($\sqrt{s_{NN}} = 39$ GeV)

- Invariant yields of pions ($\pi^{\pm}$), kaons ($K^{\pm}$), and protons ($p/\bar{p}$) at $|y| < 0.5$.
- **Mass-dependent slope separation:** The pion spectrum is steep, while the proton spectrum is flatter at low p_T.
- This mass-dependent slope flattening is a signature of **radial flow** boosting heavier hadrons.

Interactive Spectra Analyzer

Open in your browser:

- `animations/09_pt_mt_spectra_analyzer.html`

  (Observe slope flattening and p_T vs. m_T straight lines)

Collective Radial Flow: Expansion Mechanism

Fireball Pressure Gradient (Sahoo §5.2.8)

The extreme energy density created in heavy-ion collisions produces a hot, dense fireball. The high pressure gradient between the center of the fireball and the surrounding vacuum drives a collective outward radial expansion with velocity $\mathbf{v}_{flow}$.

Velocity Superposition

The velocity of a thermalized particle at freeze-out is a superposition of the collective expansion velocity and its random thermal motion:

$$\mathbf{v}_{\text{total}} = \mathbf{v}_{flow} + \mathbf{v}_{th}$$

Collective Radial Flow: Mass-Dependent Slopes

Extracted Temperature Superposition (Sahoo Eq. 5.222)

For low transverse momenta, the average kinetic energy consists of a thermal term and a collective mass-dependent term:

$$\langle K.E.\rangle = \frac{3}{2}k_B T_{th} + \frac{1}{2}m_0 v_{flow}^2$$

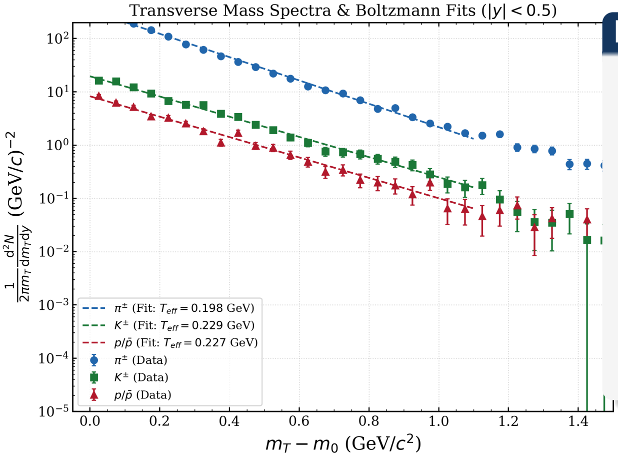
This manifests as a mass-dependent effective slope parameter T_{eff}:

$$T_{eff} = T_{th} + \frac{1}{2}m_0 \langle v_{flow}\rangle^2$$

For highly relativistic particles or high-p_T limits, the relativistic Doppler formula is used (Sahoo Eq. 5.224):

$$T_{eff} = T_{th}\sqrt{\frac{1+v_{flow}}{1-v_{flow}}}$$

AMPT: Transverse Mass spectra & Boltzmann fits



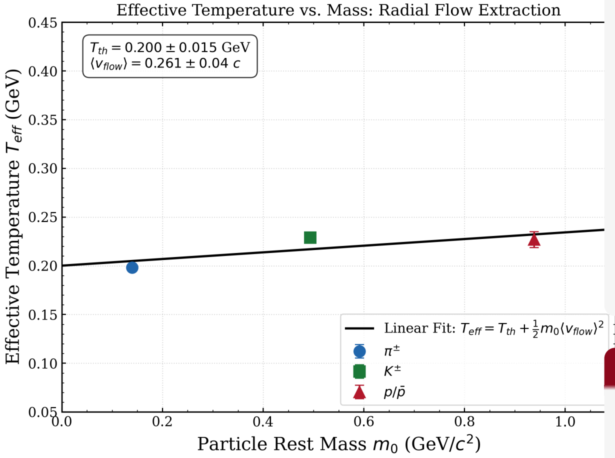
Linearization on Logarithmic Plot

- Plotting invariant yield vs. $m_T - m_0$ shows straight lines for $m_T - m_0 < 1.0$ GeV.
- Slope of each straight line is equal to $-1/T_{eff}$:

$$\ln(\text{Yield}) \propto -\frac{1}{T_{eff}}(m_T - m_0)$$

- **Effective Temperature extraction:**
  - Pions ($\pi^\pm$): $T_{eff} \approx 114$ MeV
  - Kaons ($K^\pm$): $T_{eff} \approx 139$ MeV
  - Protons ($p/\bar{p}$): $T_{eff} \approx 175$ MeV

Radial Flow Extraction: T_{eff} vs. Mass



Linear Slope Fit Method

Plotting the extracted effective temperatures T_{eff} vs. rest mass m_0 tests the linear flow model:

$$T_{eff}(m_0) = T_{th} + \frac{1}{2} m_0 \langle v_{flow} \rangle^2$$

A linear regression yields:

- **Intercept = Thermal Temp (T_{th}):** ≈ 103 MeV. The kinetic freeze-out temperature when elastic scatterings cease.
- **Slope $\Longrightarrow$ Radial Velocity ($\langle v_{flow} \rangle$):** ≈ 0.39 c.

Radial Flow Simulator

Open `animations/09_pt_mt_spectra_analyzer.html` in your browser to sweep the radial velocity slider.

Radial Flow: ZPC Partonic Stage in AMPT

ZPC Parton Scattering Knob

In AMPT, changing the ZPC parton scattering cross-section (σ_{parton}) directly alters collectivity:

- **Larger** σ_{parton}: Builds higher pressure gradients in the partonic phase, generating a larger collective radial expansion velocity $\langle v_{flow} \rangle$.
- **Smaller** σ_{parton}: Leads to a more weakly interacting medium (closer to free expansion), reducing the extracted $\langle v_{flow} \rangle$.

Parton Cascade Dynamics

The ZPC phase represents the key partonic scattering phase of the AMPT model. Higher interaction rates during this stage increase the system's anisotropic and isotropic push.

Radial Flow: ART Hadronic Freeze-Out in AMPT

ART Hadron Freeze-Out

The collective push and high density keep the system in local thermal equilibrium longer:

- This delays the freeze-out process in the ART hadron transport phase.
- As a result, the system expands and cools for a longer duration, leading to a **lower thermal freeze-out temperature** T_{th}.
- We thus observe an anti-correlation: larger collectivity builds higher $\langle v_{flow} \rangle$ but leads to a lower T_{th}.

$\langle p_T \rangle$ vs. Multiplicity: Van Hove Signature

Probing the QCD Equation of State (Sahoo pg. 164)

Leon Van Hove proposed that studying the event-by-event average transverse momentum $\langle p_T \rangle$ as a function of the charged particle multiplicity density $\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta$ would reveal the QCD phase transition:

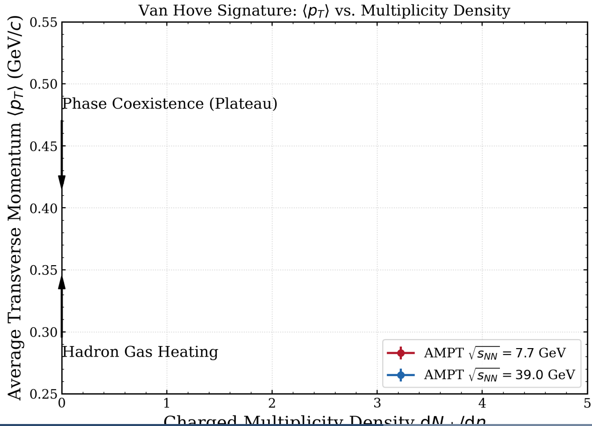
- $\langle p_T \rangle$ acts as a proxy for the **temperature** T of the early fireball.
- $\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta$ acts as a proxy for the **entropy density** s.

First-Order Phase Transition Behavior (Boiling analogy)

Like heating water, we expect three stages in a first-order phase transition:

1. **Hadronic heating:** Temperature T and $\langle p_T \rangle$ rise quickly.
2. **Mixed-phase plateau:** Multiplicity density increases as latent heat is absorbed to melt hadrons into quarks/gluons, but temperature remains flat at T_c.
3. **QGP heating:** Once all hadrons have melted, further energy input heats the pure QGP phase, causing $\langle p_T \rangle$ to rise again.

AMPT: The Van Hove Phase Transition Signature



Observations in AMPT default model

- **Low multiplicity**: Rapid rise of $\langle p_T \rangle$ indicating hadronic gas warming and expanding.
- **High multiplicity**: Clear flattening or plateau at $\langle p_T \rangle \approx 0.41$ GeV/c.

Do we see a Phase Transition in AMPT?

Standard AMPT with string melting does not contain an explicit first-order phase transition Equation of State (EoS).

- However, modified versions incorporating a **Chiral Quark-Meson Model** (CQMF) or explicit first-order phase transition EoS show a distinct flat plateau in $\langle p_T \rangle$ due to latent heat absorption.

Summary: Day 3 Hands-On Laboratory Exercises

Objective: Invariant Yields, Boltzmann Fitting & Radial Flow

All exercises are located in your Jupyter Notebooks in the `exercises/` folder:

- `exercises/day03_hands_on.ipynb` (Student skeleton notebook)
- `exercises/day03_hands_on_solutions.ipynb` (Completed reference solution notebook)

1. **Problem 1:** Calculate collider luminosity and runtime requirements for LHC *pp* runs.
2. **Problem 2:** Compute Feynman scaling variable x_F for charged pions and plot its distribution.
3. **Problem 3:** Parse AMPT events, extract identified particles (π, K, p), and construct invariant yields vs. p_T.
4. **Problem 4:** Fit transverse mass spectra $(m_T - m_0)$ to Boltzmann distributions, extract effective temperatures, and linear-fit T_{eff} vs. mass to extract T_{th} and $\langle v_{flow} \rangle$.
5. **Problem 5:** Correlate event-by-event $\langle p_T \rangle$ with multiplicity density to extract the Van Hove signature at 7.7 and 39 GeV.